

AI Meeting Assistant: Model Architecture and Data Flow

Technical Report

Overview

The AI Meeting Assistant employs a sequential, multi-stage processing pipeline that leverages a specialized speech-to-text (STT) model and two distinct language model (LLM) stages to convert raw audio into structured, actionable meeting records.

Speech-to-Text Model: Deepgram Nova-2

Role: The Deepgram Nova-2 model operates at the acoustic and phonetic layer. Its primary responsibility is to transcribe raw audio into text with high accuracy. It provides millisecond timestamps, word-level confidence scores, and applies deep-learning-based speaker diarization to differentiate speakers. It efficiently handles multi-accent speech, domain-specific vocabularies, and background noise filtering.

Language Model Stage 1: Groq GPT OSS (Transcript Refinement)

Role: The first LLM stage uses Groq GPT OSS for domain vocabulary refinement and cleanup. It corrects acoustic misrecognitions using semantic context and meeting subject cues. Furthermore, it cleans up speech disfluencies (such as fillers, false starts, and stuttering) while preserving the conversational meaning, and formats the conversation flow with clear speaker tags and cohesive paragraphs.

Language Model Stage 2: Groq GPT OSS (Minutes & Decisions)

Role: The second LLM stage also utilizes Groq GPT OSS, focusing on high-level semantic analysis and synthesis. It transforms the refined transcript into an executive summary and agenda breakdown. It extracts consensus points, formal agreements, architectural/business decisions, and concrete action items (including assignees, deadlines, and dependencies). The output is strictly formatted into human-readable Markdown and schema-validated JSON.

Data Flow Between Stages

The outputs move through the pipeline in a strictly decoupled, sequential manner:

1. **Audio Input:** Raw audio files (MP3, WAV, M4A, etc.) are fed into the system.
2. **Stage 0 → Stage 1 (Raw Transcript):** Deepgram Nova-2 processes the audio and outputs a *Raw Transcript* (diarized, but containing potential acoustic errors and speech disfluencies).
3. **Stage 1 → Stage 2 (Refined Transcript):** The Raw Transcript is ingested by Groq LLM Stage 1. This stage processes the text and produces a *Refined Transcript*, which is a polished, grammatically corrected version with clear speaker formatting.
4. **Stage 2 → Final Output (Structured Data):** Finally, the Refined Transcript is passed to Groq LLM Stage 2, which synthesizes the information into *Structured Outputs* (Markdown MoM and JSON) for interactive display, export, and API ingestion.